

Homivis Avatar Studio

Technical Project Report — Sessions 1–10

What we built · What worked · What failed · Fixes applied · Recommendations

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Stack	FastAPI · Python · ElevenLabs · Groq · fal.ai
Author	homivisuk@gmail.com
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1. Project Overview

COMFi is an AI-powered personal brand platform built under the working name **Homivis Avatar Studio**. The platform allows creators to build a digital AI persona — complete with a consistent visual identity, synthetic voice, scripted podcast-style video, and social-ready content — with minimal manual effort.

The system is designed for solo creators and micro-influencers who want a professional broadcast presence without a production team. A user describes their persona once (appearance, voice, personality, expertise), and the platform generates everything: portrait photography, spoken-word audio, talking-head video, and a fully branded content workflow.

1.1 Core Vision

The guiding principle is **consistency at scale** — every output (portrait, voice, video) should look and sound unmistakably like the same person, session after session, without the creator having to re-specify who they are. This is achieved through:

- A **persona skill file** — a structured text document capturing appearance, voice profile, expertise topics, and brand tone.
- A **locked master portrait** — a single reference face image uploaded to fal CDN and used as the identity anchor for all subsequent generations.
- A **consistent scene prompt** — a fixed set of cinematic descriptors (podcast armchair, microphone, warm lighting, red outfit) applied to every portrait.
- A **persistent voice clone** — an ElevenLabs voice ID stored on the persona and reused for every audio generation.

1.2 The Reference Persona — Vivienne

Throughout development, the primary test persona has been **Vivienne** — a 28-year-old British-Nigerian digital wellness and lifestyle creator. Vivienne is styled to match the aesthetic of reference accounts *@linakodi1_* and *@mayaa_speaks* on Instagram: natural curly hair, warm editorial lighting, confident seated poses, rich deep skin tones. All portrait and video pipeline decisions were validated against how well Vivienne's outputs matched this reference aesthetic.

2. Architecture & Technology Stack

The application is a single-process Python backend deployed on Railway, with a vanilla JavaScript single-page frontend served as a static file. All AI workloads are offloaded to external APIs — the server itself never runs a model.

Layer	Technology	Role
Backend	FastAPI (Python)	REST API, job orchestration, file serving
Frontend	Vanilla JS / HTML / CSS	SPA — 3-stage wizard UI
Deployment	Railway	Auto-deploy from GitHub push
Image generation	fal.ai — instant-character, FLUX Dev	Portrait generation with face lock
Video generation	fal.ai — Latent Sync	Talking-head lip-sync video
Voice synthesis	ElevenLabs	Voice cloning & TTS
LLM / Script	Groq (Qwen 3)	Script writing, persona parsing
Background removal	rembg (BRIA model)	Portrait compositing
Face similarity	InsightFace buffalo_l	Identity score quality check
Storage (temp)	Railway container disk	Job files, portrait images
Storage (CDN)	fal.ai CDN	Master portrait URLs (persists across redeloys)
CI/CD	GitHub → Railway webhook	Every push to main triggers redeploy

2.1 Data Flow — Portrait Generation

The portrait generation pipeline follows this sequence:

Step	Action
1	User opens Quick Portrait tab, selects persona from dropdown
2	Frontend sends POST /generate-portrait with persona_id, appearance, outfit, master_url
3	Backend resolves master_url: rejects local paths, checks MASTER_PORTTRAITS dict, case-insensitive name match
4	If master URL found → fal-ai/instant-character (face-locked, scene-following)
5	If no master → fal-ai/flux/dev (new face, scene-following)
6	fal returns CDN image URL → backend downloads to disk → serves as /portrait/{jid}/{idx}
7	Frontend displays image, shows method badge (Identity locked / FLUX fallback / FLUX new face)
8	User can: Lock as master (calls /set-master-portrait) · Use for video · Discard

3. What We Built — Feature Inventory

Feature	Status	Description
Persona Skill Parser	WORKING	Upload a .txt skill file → Groq/Qwen3 extracts structured persona data (name, age, appearance)
Persona Management UI	WORKING	Create, edit, select active persona. Persona list stored in localStorage. Dropdown pre-selects c
Master Portrait Locking	WORKING	Upload best portrait → POST /set-master-portrait → uploaded to fal CDN → stored in MASTER
Quick Portrait — FLUX Dev	WORKING	When no master portrait is set, generate a baseline seated portrait using FLUX Dev. Good scen
Quick Portrait — instant-character	WORKING	When master portrait is set, use fal-ai/instant-character for face-consistent seated portrait gener
Portrait Quality Score	WORKING	InsightFace buffalo_l cosine similarity between generated portrait and master. Score displayed
Background Removal	WORKING	rembg BRIA model removes portrait background for compositing. Pre-warmed on startup to elimin
Script Generation	WORKING	POST /generate-script with topic + persona → Groq/Qwen3 generates a 200-250 word podcast
Voice Synthesis	WORKING	POST /generate-voice with script + persona voice_id → ElevenLabs TTS → returns audio file. V
Flashtalk Video	WORKING	POST /generate-video: composites portrait onto background, runs Latent Sync lip-sync, returns
Scene Generator	WORKING	POST /generate-scene — standalone scene background generation via FLUX Schnell for comp
3-Stage UI Wizard	WORKING	Stage 1 (Portraits), Stage 2 (Script & Voice), Stage 3 (Video). Each stage gated on prior stage c
Static Asset Serving	WORKING	FastAPI mounts /assets static route for CSS/JS. /portrait/{jid}/{idx} serves generated images.
Health Endpoint	WORKING	GET /health returns version and status. Used by Railway for uptime monitoring.
TikTok/Instagram Publishing	PENDING	Direct social publishing via platform APIs. Not yet implemented.
Cloudflare R2 Storage	PENDING	Move video/portrait storage from Railway ephemeral disk to R2. Not yet implemented.
Caption Generation	PENDING	Auto-generate captions from script and burn onto video. Not yet implemented.
Video Aspect Ratio Selector	PENDING	UI selector for 9:16 (TikTok), 1:1 (Instagram), 16:9 (YouTube). Backend wired, UI not connecte

4. What Worked

4.1 instant-character for Face-Consistent Portraits

After evaluating multiple fal.ai models (see Section 7), **fal-ai/instant-character** proved to be the right solution for the core use case: generating a seated, scene-following portrait of a specific person using a single reference image. Once a valid fal CDN URL is stored as the master portrait, instant-character reliably:

- Locks the face identity to the reference image at scale=0.8
- Follows complex scene prompts (seated armchair, microphone, background plant, red outfit, studio lighting)
- Returns portrait-orientation images at portrait_4_3 aspect ratio
- Costs approximately \$0.06 per generation — acceptable for a creator tool
- Runs in ~25–35 seconds — within user tolerance for a single portrait

4.2 Railway + GitHub CI/CD Pipeline

The deploy pipeline proved very reliable throughout development. Every git push to the main branch automatically triggers a Railway build and redeploy, typically completing within 2–3 minutes. This enabled rapid iteration: fix a bug in the code → push → test within minutes. Railway's log streaming provided immediate visibility into server-side errors.

4.3 Groq + Qwen3 for Script Generation

Script generation using Groq's Qwen 3 model performed well. The model reliably produces 200–250 word podcast intro scripts that match the persona's voice, tone, and area of expertise as specified in the skill file. Latency is low (~2–3 seconds) and cost is negligible.

4.4 ElevenLabs Voice Cloning

Voice generation using ElevenLabs and a pre-cloned voice ID has been consistent and high-quality. Once a voice is cloned and its ID stored on the persona, all subsequent audio generations use the same voice without any re-upload or re-cloning. The ElevenLabs API is stable and fast (~5–8 seconds for a 250-word script).

4.5 Latent Sync Lip-Sync Video

fal-ai/latentsync for talking-head video generation works reliably once the portrait and audio are properly prepared. The compositing pipeline (rembg background removal → background image → portrait overlay → Latent Sync) produces clean, broadcast-ready talking-head video. The main remaining issue is aspect ratio (currently portrait-only).

4.6 rembg Background Removal

The BRIA model via rembg consistently produces clean, accurate background removal on portrait images. The pre-warming on server startup (spinning up the model once at boot) eliminates the cold-start latency that previously caused timeouts on the first request.

5. What Did Not Work — Issues Log

The following table documents every significant failure encountered during the project, its root cause, and the resolution applied.

Issue	Status	Resolution / Note
"Persona not found" on Quick Portrait	FIXED	populateQpPersonaSelect() stored p.id as dropdown value, but personas have n
Double /portrait/ path in image URL	FIXED	Frontend built API()+'/portrait/'+d.images[0] but d.images[0] already equals '/portr
PuLID guidance_scale validation error	FIXED	Sent guidance_scale=7.0 and num_inference_steps=20 to PuLID. PuLID limits a
PuLID ignores scene prompt — headshot only	FIXED	PuLID is a headshot/face-swap model. Even with scene text in the prompt, output
FLUX Dev image_prompt parameter does not	FIXED	Attempted to pass image_prompt and image_prompt_strength to FLUX Dev for fa
FLUX Schnell ignores scene prompts	FIXED	At 8 inference steps, FLUX Schnell produces generic portraits and ignores compl
Seated portrait — face not locked to Vivienne	PARTIAL	instant-character is running correctly, but MASTER_PORTTRAITS is an in-memor
Railway upstream error / request timeout	FIXED	fal_client.subscribe() is synchronous and blocks for 25-35s. Running it directly in
Method badge showing wrong model	FIXED	Response JSON always returned method:'pulid' (hardcoded string). The actual m
rembg cold-start timeout	FIXED	First request after deploy caused rembg to download the BRIA model weights (~5
Static files not served from /assets	FIXED	FastAPI StaticFiles mount path was wrong on initial deploy. Fixed mount point an
Video aspect ratio — face-only crop	PENDING	Latent Sync output is portrait-oriented but the subject's face fills most of the fram
MASTER_PORTTRAITS lost on redeploy	PARTIAL	In-memory dict is cleared every Railway redeploy. fal CDN URLs are durable but

6. Solutions Implemented

6.1 Persona Key Fix — Name as Identifier

The root cause of the "Persona not found" error was using `p.id` as the dropdown option value. Personas in the system have no `.id` field — name is the unique key. The fix was applied consistently across every persona-related function in the frontend:

```
// BEFORE (broken – p.id is undefined)
opt.value = p.id;

// AFTER (correct – name is the key)
opt.value = p.name;

// All downstream lookups updated to match:
const p = S.personas.find(x => x.name === pid);
```

6.2 Portrait URL Fix

The backend returns image paths as `/portrait/{jid}/{idx}` in the images array. The frontend was incorrectly prepending `/portrait/` again, resulting in a double path. Fix:

```
// BEFORE (double path – 404)
img.src = API() + '/portrait/' + d.images[0];

// AFTER (d.images[0] already contains /portrait/jid/0)
img.src = API() + d.images[0];
```

6.3 Thread Pool for fal.ai Calls

This was the most impactful fix of Session 10. All `fal_client.subscribe()` calls are synchronous and blocking — they hold for the full generation time (25–35s for instant-character). Running them directly in an async FastAPI route handler blocks the entire asyncio event loop and hits Railway's request proxy timeout. The fix moves all fal calls into a dedicated thread via `run_in_executor`:

```
loop = asyncio.get_event_loop()

def _run_fal_generation():
    # All fal_client.subscribe() calls here – blocking but off the event loop
    result = fal_client.subscribe('fal-ai/instant-character', arguments={...})
    return images, scores, method

# In the async handler:
images, scores, method = await loop.run_in_executor(None, _run_fal_generation)
```

6.4 Master Portrait URL Validation

The browser stores `masterPortraitUrl` in `localStorage`, which after navigating away may be a local Railway path like `/portrait/abc123/0` rather than a public fal CDN URL. `fal.ai` models require a publicly-accessible HTTPS URL for the reference image. The backend now validates and falls back correctly:

```
ref_url = master_url.strip()
# Reject local paths – fal cannot fetch them
if ref_url and not ref_url.startswith('http'):
```

```
    ref_url = ''
# Fall back to in-memory store (populated via /set-master-portrait)
if not ref_url and persona_id:
    entry = MASTER_PORTRAITS.get(persona_id)
    if entry: ref_url = entry['url']
# Case-insensitive name match as final fallback
if not ref_url and persona_id:
    for key, entry in MASTER_PORTRAITS.items():
        if key.lower() == persona_id.lower():
            ref_url = entry['url']; break
```

7. AI Model Research & Selection

Selecting the right fal.ai model for face-consistent, scene-aware portrait generation was a central challenge. The following models were evaluated:

Model	Face Lock	Scene Prompt	Cost	Verdict
fal-ai/pulid	YES — tight	NO — headshot only	~\$0.04	Rejected: no scene control
fal-ai/flux/schnell	NO	POOR at 8 steps	~\$0.01	Rejected: ignores scene
fal-ai/flux/dev	NO	GOOD at 28 steps	~\$0.025	Used for no-master baseline
fal-ai/flux/dev (image_prompt)	NO	N/A	N/A	Rejected: param doesn't exist
fal-ai/instant-character	YES — flexible	EXCELLENT	~\$0.06	SELECTED: primary model

7.1 Why instant-character Was the Right Choice

fal-ai/instant-character is built specifically for the use case this project requires: generating multiple images of the same character in different scenes using a single reference photo. Unlike PuLID (which does face-swapping into tight headshots only) and IP-Adapter variants (which are weak at scene following), instant-character uses a character-consistency architecture that locks identity while leaving the diffusion model free to follow the scene prompt.

Key parameters used in production:

```
fal_client.subscribe('fal-ai/instant-character', arguments={
    'prompt': prompt,           # Full scene description
    'image_url': ref_url,       # fal CDN URL of master portrait
    'scale': 0.8,               # Face identity strength (0=none, 2=max)
    'guidance_scale': 3.5,     # Prompt adherence
    'num_inference_steps': 28, # Quality steps
    'image_size': 'portrait_4_3',
    'num_images': 1
})
```

8. Recommendations

The following recommendations are prioritised by impact. High-priority items address reliability and data loss risks. Medium items improve the product experience. Low items are future growth features.

Priority	Recommendation	Effort	Impact
HIGH	Persist MASTER_PORTRAITS to disk / Cloudflare R2	Medium	Critical reliability gain — prevents face
HIGH	Show 'Master portrait not set' warning if MASTER_PORTRAITS is empty	Low	UX clarity — user knows why face isn't
HIGH	Move video storage to Cloudflare R2 (free 10 GB)	Medium	Railway ephemeral disk means all gen
HIGH	Add /reload-masters endpoint to re-register persisted CDN URLs on deploy	Low	Self-healing after redeploy — server re
MEDIUM	Fix Flashtalk video to show upper body (landscape 768x448 output)	Medium	Current output crops to face only — ru
MEDIUM	Wire aspect ratio UI selector to backend (9:16, 1:1, 16:9)	Low	Enable TikTok (9:16), Instagram squar
MEDIUM	Add auto-caption generation from script to video output	Medium	Critical for TikTok/Instagram Reels —
MEDIUM	Rate-limit Quick Portrait to 1 in-flight job per persona	Low	Prevents double-spend if user double-
MEDIUM	Show generation progress indicator (spinner + elapsed time)	Low	instant-character takes 25-35s — user
LOW	Implement TikTok publishing API integration	High	One-click publish from the Studio to Ti
LOW	Implement Instagram Reels API integration	High	Same as TikTok — completes the cont
LOW	Add video gallery — persist all generated videos per persona	Medium	Currently videos are ephemeral — no
LOW	Scale to multi-tenant — users register their own ElevenLabs/fal API keys	High	Required for public launch — currently
LOW	Add Stripe billing and usage metering	High	Prerequisite for commercialisation — t

9. Immediate Next Steps

The following actions should be completed in the next session to stabilise the current build and unlock the next phase of the product:

#	Priority	Action	Detail
1	HIGH	Re-lock Vivienne master portrait	Go to Stage 1 → Portraits. Click Generate. Once you have a good portrait, click 'Set as master portrait'.
2	HIGH	Persist MASTER_PORTTRAITS	Write MASTER_PORTTRAITS to /app/uploads/master_portraits.json on every set. Load on next session.
3	HIGH	Fix video aspect ratio for upper half	Change Latent Sync output to 768x448 landscape or add explicit framing to the portrait video.
4	MEDIUM	Add generation progress UI	Show a spinner with elapsed seconds during portrait generation. instant-character takes ~10s.
5	MEDIUM	Connect aspect ratio selector	The backend already supports different sizes. Wire the frontend radio buttons to send the selected size.
6	LOW	Add caption overlay to video	After Latent Sync generates the video, use ffmpeg to burn subtitles from the script as vtt.

10. Appendix — API Reference

Quick reference for all backend endpoints.

Method	Endpoint	Description
GET	/health	Health check. Returns version and status.
GET	/	Serves index.html SPA.
POST	/parse-skill	Parse persona skill file text → structured JSON.
POST	/generate-script	Generate podcast script. Params: persona_name, topic, persona_skill.
POST	/generate-voice	Generate audio from script. Params: script, voice_id, persona_name.
POST	/set-master-portrait	Upload master portrait → stores to fal CDN. Params: persona_id (form), portrait (file).
POST	/get-master-portrait	Return stored master URL for persona. Params: persona_id.
POST	/generate-portrait	Generate portrait (instant-character or FLUX Dev). Params: persona_name, persona_skill.
GET	/portrait/{jid}/{idx}	Serve generated portrait image file.
POST	/generate-scene	Generate background scene image via FLUX Schnell.
POST	/generate-video	Full video pipeline: script + voice + portrait → Latent Sync talking-head .mp4.
GET	/video/{jid}	Stream generated video file.
POST	/remove-background	Remove background from uploaded image using rembg BRIA.